Day 19: Retrieval-Augmented Generation (RAG) Fundamentals

1. What is RAG and Why Does It Beat a Bare LLM?

RAG (Retrieval-Augmented Generation) is an architecture that optimizes the output of a Large Language Model by referencing an authoritative, external knowledge base before generating a response

A bare LLM is prone to **hallucinations** and has a fixed knowledge cutoff date.

RAG solves this by ensuring the LLM only answers questions using real, retrieved document context, making the AI highly reliable for business automation

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2. The Ingestion Path Diagram

[Raw PDFs/Docs] —–> [Extract Text] —–> [Split into Chunks] —–> [Generate Vector Embeddings] —–> [Store in Pinecone DB

3. The Query Path Diagram

[User Question] —–> [Vector Search DB] —–> [Retrieve Matching Chunks] —–> [Feed Context + Question to LLM] —–> [Factual Answer]

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4. Why Chunk Size Matters

Chunking is the process of breaking long documents into smaller, coherent text blocks before generating vector embeddings

- **Too Large:** Floods the LLM's context window with noisy, irrelevant data, diluting accuracy .
- **Too Small:** Strips out vital surrounding context, preventing the LLM from understanding the complete meaning of the passage
- **Standard Best Practice:** Chunk sizes of 500 to 1,000 characters with a 10% overlap to preserve context boundaries